

Characterizing Extended Kalman Filter Fusion of Visual Detection and Odometry Under Sparse Observation: A Multi-Robot Digital Twin Case Study

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Abstract. Multi-robot systems combining odometry with external visual detection often rely on a discrete fallback — trusting detection when available, odometry otherwise — which discards uncertainty information and offers no principled behavior during long detection gaps. We replace this fallback with a per-robot Extended Kalman Filter fusing odometry and YOLO-based visual detection from a fixed overhead camera, validated in two stages (synthetic NEES/NIS consistency, then real recorded data) on a ROS2/Gazebo digital twin using three TurtleBot3 robots. Measured at correction instants, the filter reduces position error by +23.7% relative to odometry alone under aggressive drift. However, measured continuously — at every odometry reading — the same filter is 12.7% *worse* than raw odometry, under the sparse visual coverage (0–28%) our camera setup exhibits in practice. We show this reversal is the expected consequence of operating below a critical observation rate, not a design flaw, and connect it mechanistically to a covariance ceiling required to keep Mahalanobis-gated association stable. We argue that reporting both metrics, rather than only the commonly reported correction-instant gain, is necessary to honestly characterize EKF fusion under sparse observation, and discuss why alternative filters (UKF, particle filter, IMM) do not resolve the underlying limitation.

Keywords: Extended Kalman Filter · Sensor Fusion · Multi-Robot Localization · Sparse Observations · Digital Twin

1 Introduction

Multi-robot systems that combine onboard odometry with external visual detection face a recurring design choice: how to combine two position estimates that are individually noisy, arrive at different rates, and are not always both available at the same instant. A common and pragmatic solution — used, for instance, in our own CERISE multi-robot digital twin prior to this work — is a *discrete fallback*: use the visual detection when one is confidently associated to a robot, and fall back to raw odometry otherwise. This is simple and robust, but it discards information: it does not let a low-confidence detection partially

correct odometry, does not propagate uncertainty from either source into the estimate, and provides no principled way to reason about what happens during long stretches without any visual detection at all.

Extended Kalman Filtering (EKF) is the classical answer to this class of problem, offering a full probabilistic treatment of prediction and correction with explicit uncertainty propagation. Its convergence properties, however, are classically established under the assumption of reasonably dense, regular observation [2] — an assumption that a single fixed overhead camera, subject to a finite field of view and self-occlusion during robot maneuvers, does not generally satisfy. Whether and how this matters in practice, on a real robotic platform rather than only in simulation, is the question this paper investigates.

We replace CERISE’s discrete fallback with a per-robot EKF fusing odometry and YOLO-based visual detection from an overhead camera, and evaluate it end-to-end on a ROS2/Gazebo digital twin using three TurtleBot3 robots. Our contribution is twofold. First, we validate the filter in two stages — synthetic NEES/NIS consistency, then real recorded data — and report a real fusion gain (+23.7% position error reduction relative to odometry alone, under aggressive drift, measured at the instants a visual correction occurs). Second, and more centrally, we show that the same filter, measured continuously rather than only at correction instants, is 12.7% *worse* than raw odometry under the sparse visual coverage (0–28%) our overhead-camera setup actually exhibits — and we explain this reversal mechanistically, connecting it to established theory on Kalman filtering under intermittent observations [3,4]. We argue that reporting both metrics, rather than the more commonly reported correction-instant gain alone, is necessary for an honest characterization of EKF fusion in this regime, and we discuss why alternative filters do not resolve the underlying limitation.

2 Related Work

Our work builds on three layers of prior literature: (i) the classical formulation and convergence theory of EKF-based localization, (ii) the theory of Kalman filtering under intermittent or sparse observations, and (iii) recent multi-robot systems that fuse odometry with visual detection in realistic simulation or physical testbeds.

2.1 Classical EKF Localization and Convergence

Bailey and Durrant-Whyte [1] provide the standard tutorial formulation of EKF-based simultaneous localization and mapping, formalizing the motion model $x_k = f(x_{k-1}, u_k) + w_k$ and observation model $z_k = h(x_k) + v_k$ that our filter follows directly: odometry drives prediction, and YOLO-based visual detections drive correction. We adopt this notation throughout Section 3.

Dissanayake et al. [2] prove convergence properties of EKF-SLAM covariance under the assumption of *dense* landmark observation — i.e., that the filter is corrected regularly and predictably. This assumption is central to our Discussion

(Section 6): our overhead-camera setup violates it by design, since a robot is visually detected in only 0–28% of odometry readings depending on scenario, and this violation is precisely what produces the continuous-error finding we report.

2.2 Kalman Filtering Under Intermittent Observations

Sinopoli et al. [3] establish that, when observations arrive as a Bernoulli process, there exists a critical arrival probability below which the expected estimation error covariance diverges. Our Scenario 3 (curve/occlusion, 0% visual correction over the full recording) sits at the limiting case $p \rightarrow 0$ covered by this theory — the covariance saturation we observe is therefore an expected consequence of the observation regime, not a symptom of filter misconfiguration.

Censi [4] revisits Sinopoli et al.’s result and shows that the existence and value of this critical threshold depend on the error metric used to summarize the covariance matrix (Euclidean vs. Riemannian manifold of SPD matrices). We report divergence using $\text{trace}(P)$ (Section 5), and note this metric-dependence explicitly as a scoping remark rather than a claim of a universal threshold.

2.3 Recent Multi-Robot Visual-Odometric Fusion and Positioning

Khosravi et al. [7] report a decentralized cooperative localization scheme for two robots, fusing local EKF estimates with intermittent mutual LiDAR corrections (Gazebo and physical hardware), with a 34–56% RMSE reduction from fusion — the same order of magnitude as our +23.7% correction-instant gain. Ahmed and Frémont [8] fuse a custom YOLOv8-RGBDis detector with a per-target EKF for decentralized tracking of a heterogeneous team (three UAVs, one surface vessel) in a maritime Gazebo scenario, reporting $\text{mAP}@0.5 = 0.995$ for their detector — close to our own 0.995 despite a different architecture and domain, a coincidence we note but do not lean on. Neither work reports a continuous-error regime in which fusion underperforms the unfused baseline — the central, differentiating finding of our work. Dobrynin and Garcia [9] share our exact platform (three TurtleBot3 units, ROS2) but solve camera-free relative localization, and so do not encounter the sparse visual coverage regime we characterize; we treat this as complementary work, revisited in Section 7.

Our contribution is therefore the joint report of both the correction-instant gain and the continuous-error degradation under sparse coverage, together with a theoretical grounding ([3,4]) for why this trade-off is expected rather than a bug in our implementation.

3 Methodology

Following the standard EKF formulation of Bailey and Durrant-Whyte [1], each robot is tracked independently by a per-robot filter with state $x_k = [p_x, p_y, \theta]^\top$:

2D position and heading. The motion model $f(\cdot)$ integrates the *delta* of consecutive odometry readings onto the current filter state — not the absolute odometry value, which would silently discard any prior visual correction. The observation model $h(\cdot)$ maps filter state to the 2D position reported by a YOLO-based robot detector observing the scene from a fixed overhead camera (Section 4).

3.1 Prediction: Process Noise and Covariance Capping

Process noise is a diagonal matrix $Q = \text{diag}(0.5, 0.5, 0.25)$, scaled by $\Delta t/0.1$ at each prediction step to correct for the real odometry rate (≈ 29 Hz, $\Delta t \approx 0.034$ s) differing from the 0.1 s assumed during synthetic calibration (Section 4). Q was calibrated empirically against real recorded data, not only the synthetic NEES/NIS procedure: a sweep from 10^{-6} to 1.0 showed a monotonically increasing fusion gain up to saturation between $Q = 0.5$ and $Q = 1.0$.

A hard ceiling is applied to the covariance matrix after each prediction, $\text{clip}(P, -\infty, 0.05)$, applied element-wise. Without this ceiling, P was observed to grow unbounded during long intervals without visual correction (reaching $\text{trace}(P) \approx 85$ in one recorded event, against an expected $\text{trace}(P) \approx 0.5$), which in turn breaks Mahalanobis-gated association (Section 3.3): an inflated covariance widens a robot’s acceptance region enough to incorrectly claim a neighboring robot’s detection. This capping strategy is functionally analogous to *track coasting* in classical multi-target tracking (Bar-Shalom), and is theoretically motivated by Sinopoli et al.’s [3] result that covariance diverges below a critical observation arrival rate — capping trades strict optimality for bounded, predictable behavior under exactly the sparse-observation regime we operate in (Section 5). We report the covariance ceiling’s effect on continuous-time estimation error explicitly, rather than treating it as an implementation detail, in Section 5.

3.2 Correction: Confidence-Adaptive Measurement Noise

Measurement noise R is not fixed: it is derived linearly from the YOLO detection confidence $c \in [0, 1]$ (the third component of each detected pose, repurposed as a confidence channel), $\sigma = R_{\min} + (1 - c)(R_{\max} - R_{\min})$, with $R_{\min} = 0.02$, $R_{\max} = 0.20$ (meters), and $R = \text{diag}(\sigma^2, \sigma^2)$. A high-confidence detection is trusted more than a low-confidence one during the Kalman update, rather than treating all detections as equally reliable.

3.3 Data Association

Because the overhead camera observes up to three robots simultaneously, each incoming detection must be assigned to the correct robot before correction. We use Mahalanobis-gated nearest-neighbor association, computing $d^2 = e^\top P_{xy}^{-1} e$ for each (robot, detection) pair using each robot’s own 2×2 position covariance submatrix P_{xy} , and accepting the closest match only if $d^2 \leq 9.21$ (the 99%

confidence threshold of the χ^2 distribution with 2 degrees of freedom). This is stricter than raw Euclidean nearest-neighbor and accounts for each robot’s individual uncertainty, but inherits a known limitation of covariance-based gating discussed by Altendorfer and Wirkert [6]: an inflated covariance widens a track’s acceptance region, which can bias association toward lower-confidence or more uncertain tracks — precisely the failure mode that motivated the covariance ceiling described above.

3.4 Camera Calibration

Camera intrinsics are obtained via `cv2.calibrateCamera` following Zhang’s method [5], using a synthetically generated checkerboard spawned at multiple poses in the simulated scene. Reprojection error alone was found to be an unreliable calibration-quality signal: an initial calibration using five poses concentrated near the frame center achieved a deceptively low reprojection error (0.018px) while yielding a focal length $8.9\times$ larger than the value expected from the camera’s known field of view — classical overfitting from an ill-conditioned, poorly distributed pose set. We therefore add an explicit sanity check comparing calibrated f_x against the FOV-derived theoretical value, accepting only calibrations within a $0.5\times$ – $2.0\times$ ratio. The accepted calibration (four well-distributed poses) achieves $f_x = 739.4$, $f_y = 733.6$ ($1.33\times$ the theoretical ratio) with a mean reprojection error of 0.162px, within the 0–0.2px range considered good practice [5].

3.5 Two-Stage Filter Validation

Before evaluating fusion gain, we validate that the filter itself is correctly calibrated, in two independent stages:

(1) Synthetic validation via NEES/NIS. Following Bar-Shalom and Chen’s methodology, we generate a synthetic trajectory with known ground truth and run 30 independent Monte Carlo seeds (2000 steps each, 58,500 post-warmup samples). The filter achieves $\text{NEES} = 2.955$ (expected 3.0, 1.5% relative error) and $\text{NIS} = 1.982$ (expected 2.0, 0.9% relative error) — close enough to nominal values to be considered well-calibrated in practice, even though the formal χ^2 95% bounds are too narrow to pass at this sample size (an expected effect of aggregating tens of thousands of samples, not a sign of miscalibration).

(2) Real-data validation. The calibrated filter is then run against odometry and YOLO detections recorded from the live digital twin (Section 4), where — unlike the synthetic case — observation coverage is sparse and irregular by construction of the physical scenario, not by design choice.

4 Experimental Setup

We evaluate our EKF within CERISE, a ROS2/Gazebo digital twin of a multi-robot warehouse-style environment. Three TurtleBot3 Waffle units (`robot1`–`robot3`) operate in a shared simulated world alongside a static overhead camera

mounted at $(0, 0, 3)$ m with 90° downward pitch and a 60° horizontal field of view (1.047 rad), publishing 640×480 RGB frames at 10 Hz. Each robot also publishes its own odometry (`/robotN/odom`) at ≈ 29 Hz and a 2D LiDAR scan (`/robotN/scan`, unused in this work — see Section 7).

4.1 Visual Detection

Robot detection is performed by a YOLO model trained on 375 manually annotated frames captured from the same overhead camera setup, achieving $\text{mAP@0.5} = 0.995$ on the held-out validation split (image size 416px). At inference, the detector publishes a `PoseArray` on `/robot_detections`, repurposing the otherwise-unused z component of each pose to carry the YOLO confidence score, so the EKF correction step (Section 3) can consume it directly.

4.2 Recorded Scenarios

Three scenarios were recorded as ROS2 bags (MCAP format, ~ 18 s each, ~ 2 MB each), covering distinct motion and occlusion regimes:

- **Stationary** (`cenario1_parado`): robots remain at their spawn poses. 152 detection messages against 550 odometry readings per robot — the highest visual coverage of the three scenarios ($\approx 28\%$).
- **Straight motion** (`cenario2_reto`): robots move forward at $v_x = 0.1$ m/s. 144 detection messages against 522 odometry readings per robot ($\approx 11\%$ coverage); ground-truth `robot1` displacement of ~ 1.5 m confirms real motion, not merely a commanded velocity with no physical effect.
- **Curve / occlusion** (`cenario3_curva`): robots move with $v_x = 0.12$ m/s and $\omega_z = 0.4$ rad/s. 145 detection messages recorded on the topic, but **zero** of them were successfully associated to `robot1` by the gating procedure of Section 3.3 over the full ~ 18 s recording — the single fixed overhead view-point loses robots to self-occlusion and out-of-frame motion during turning, producing the 0% coverage case discussed in Section 5.

Timestamp synchronization between `/robot_detections` and odometry was verified directly via `rosvbag2_py` in all three scenarios: message-arrival windows coincide within each recording, with no gap exceeding 500 ms on any channel — ruling out sensor-clock desynchronization, a known confound in multi-sensor fusion pipelines, as an explanation for the results reported in Section 5.

4.3 Reproducibility Package

To support independent verification, we release a package containing the Gazebo world file, robot URDF (including the unused LiDAR, for completeness), camera calibration parameters, all EKF hyperparameters, and the git commit SHA corresponding to the exact code used to produce the results in Section 5. We explicitly document one non-obvious dependency: `ros-humble-rosbag2-storage-mcap` is not installed by default on ROS2 Humble and must be installed separately to read the released bags.

5 Results

We report two distinct error metrics, deliberately kept separate rather than collapsed into a single number: *correction-instant error*, measured only at the moments a YOLO detection is successfully associated and fused (the metric most commonly reported in related work — Section 2), and *continuous error*, measured at every odometry reading regardless of whether a correction occurred. As we show below, these two metrics disagree in sign, and both are necessary to honestly characterize the filter’s behavior under sparse observation.

5.1 Filter Calibration

The synthetic NEES/NIS validation described in Section 3 confirms the filter is well-calibrated before any real-data evaluation. Fig. 1 shows both statistics remaining close to their theoretical expected values (NEES $\rightarrow$ 3.0, NIS $\rightarrow$ 2.0) across the Monte Carlo run.

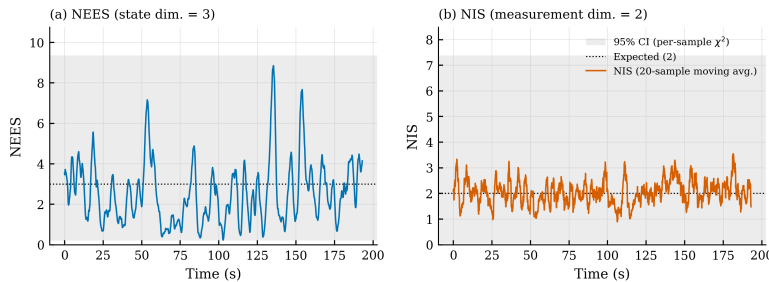


Fig. 1. NEES and NIS consistency over the synthetic Monte Carlo validation (30 seeds, 2000 steps each). Both statistics track their theoretical expected values, indicating a well-calibrated filter.

5.2 Correction-Instant Error

Table 1 reports position error under three injected odometry drift regimes (no drift, light drift, aggressive drift), aggregated across all three recorded scenarios ($n = 1323$). Under aggressive drift — the regime in which raw odometry has already accumulated substantial error before a visual correction becomes available — the EKF reduces mean position error by +23.7% relative to odometry alone (Wilcoxon signed-rank, $p < 0.001$). Under light drift, the gain is negative (−5.3%): visual detection error (≈ 3 –6 cm) is not yet smaller than the still-small odometry error, so fusion offers no benefit in this regime — expected behavior for a correctly calibrated filter, not a failure.

Table 1. Correction-instant position error (m), aggregated over 3 scenarios ($n = 1323$).

Drift regime	EKF (mean)	Odom-only (mean)	Change
None (odom = GT by construction)	0.0356	0.0000	N/A
Light	0.0528	0.0501	-5.3%
Aggressive	0.1282	0.1681	+ 23.7%

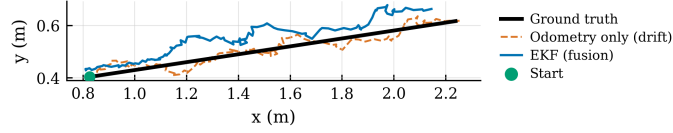
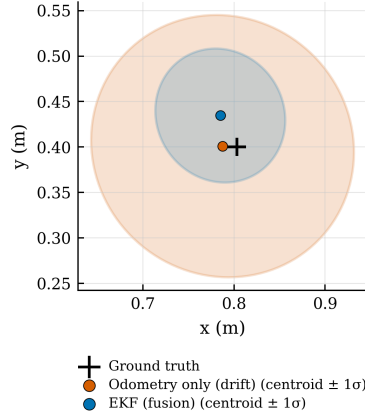
**Fig. 2.** Estimated trajectory under aggressive drift, Scenario 2 (straight motion). The EKF (blue) tracks ground truth (black) more closely than raw drifted odometry (orange, dashed).

Fig. 2 shows this qualitatively for Scenario 2: raw odometry visibly drifts away from the straight ground-truth path, while the EKF estimate tracks it more closely. For the stationary scenario, where no real displacement exists to plot as a trajectory, Fig. 3 shows the same effect as empirical dispersion: the EKF estimate cloud (blue) is visibly more compact and centered nearer ground truth than the raw odometry cloud (orange).

**Fig. 3.** Empirical 1σ dispersion ellipses (Scenario 1, stationary), computed from the sample covariance of each source's estimates around their centroid.

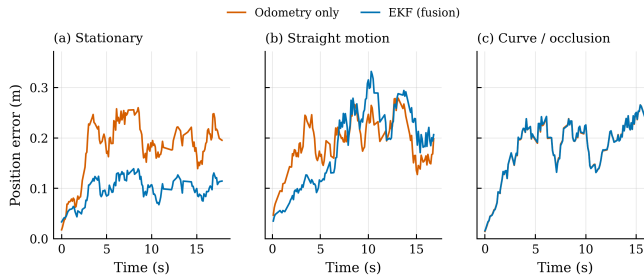


Fig. 4. Correction-instant position error over time, aggressive drift, all three scenarios.

5.3 Continuous Error: The Central Finding

When error is instead measured at *every* odometry reading — not only at correction instants — the result reverses. Table 2 reports continuous-time error under aggressive drift, using the same recordings as Table 1. Aggregated across all readings ($n = 1607$), the EKF is 12.7% *worse* than raw odometry (Wilcoxon signed-rank, $p < 0.001$) — the opposite sign from the correction-instant result.

Table 2. Continuous position error (m), every odometry reading, aggressive drift.

Scenario	YOLO coverage	EKF (mean)	Odom-only (mean)	Change
Stationary	27.6%	0.2116	0.1774	−19.3%
Straight motion	11.3%	0.2115	0.1780	−18.8%
Curve / occlusion	0.0%	0.1778	0.1778	−0.0%
Aggregated	—	0.2003	0.1777	−12.7%

Two observations explain this reversal. First, the degradation scales inversely with YOLO coverage: Scenario 3 (0% coverage — no successful visual correction over the entire ~ 18 s recording) shows a change of exactly 0.0%, because with zero corrections the EKF’s prediction step degenerates to pure odometry-delta integration, making it mathematically identical to the unfused baseline. Scenarios 1 and 2 (27.6% and 11.3% coverage) show similar $\sim 19\%$ degradation despite their different coverage levels, suggesting the effect saturates quickly once corrections become sparse enough for the covariance ceiling (Section 3) to bind.

Second, and mechanistically explaining the first, Fig. 5 shows why: $\text{trace}(P)$ saturates at the covariance ceiling almost immediately after any gap without correction, and only drops when a correction lands. In Scenario 3, the trace saturates within the first observed step and never drops again for the remainder of the recording. This is precisely the divergence regime predicted by Sinopoli et al. [3] for observation arrival rates below a critical threshold: the filter’s covariance ceiling prevents unbounded growth (and the association failures it would otherwise cause, Section 3), but it does so by capping the filter’s *own estimate of*

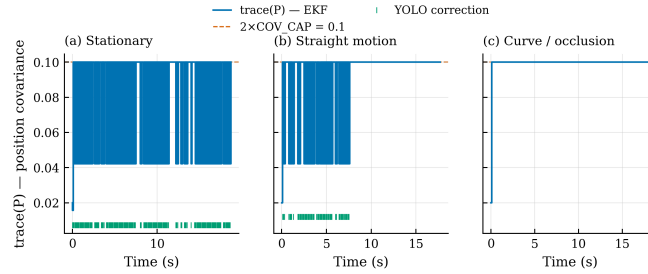


Fig. 5. $\text{trace}(P_{xy})$ over time (step plot), all three scenarios. Short vertical ticks along the bottom mark successful YOLO corrections. The dashed line marks $2 \times \text{COV_CAP} = 0.1$, the ceiling on the position-covariance trace (Section 3). Coverage decreases left to right ($27.6\% \rightarrow 11.3\% \rightarrow 0\%$), and covariance saturates at the ceiling for correspondingly longer stretches, reaching permanent saturation in the zero-coverage case.

its uncertainty, not by making its state estimate actually more accurate during the uncorrected interval — the filter remains overconfident in its odometry-only prediction relative to how uncertain that prediction actually is.

6 Discussion

6.1 The Continuous-Error Finding Is Expected, Not a Bug

The natural reading of Table 2 is unflattering: our EKF, evaluated continuously, loses to the odometry it is supposed to improve upon. We argue this is the expected consequence of operating below the critical observation rate identified by Sinopoli et al. [3], not evidence of a flawed design. Dissanayake et al.’s [2] classical convergence proof for EKF-based localization assumes regular, dense observation of landmarks; our overhead-camera setup violates that assumption by construction (0–28% coverage across scenarios), and the theory predicts exactly the covariance divergence we would see without a ceiling, and the practical consequence we do see — an overconfident filter — with one. Framed this way, Scenario 3’s 0.0% continuous-error change is not a null result but a clean empirical confirmation: with zero corrections, our filter’s prediction step is provably identical to odometry-delta integration, and the measured change is exactly zero, as the mechanism predicts it should be.

We follow Censi’s [4] caution that the existence and value of a critical threshold depends on the covariance summary metric used; we report $\text{trace}(P)$ (Fig. 5) as a scoping choice, not a claim that divergence is metric-independent.

6.2 Why EKF, and Why Covariance Capping

A natural question is whether an alternative filter would avoid this trade-off. We argue no standard Bayesian filter resolves the zero-coverage case (Scenario 3):

a particle filter degenerates to pure process propagation with no likelihood to reweight against, an Unscented Kalman Filter targets observation-model non-linearity rather than observation *rate* (our near-linear overhead projection gains little from it), and Interacting Multiple Model filters still require some observation to weight competing modes. We therefore treat the low coverage itself — a single fixed camera viewpoint subject to occlusion during turns — as the root cause, consistent with sensor-placement literature establishing that estimation limits under a given sensor topology hold independently of the filter processing its observations. The covariance ceiling (Section 3) is, in this light, an engineering approximation of *track coasting* in classical multi-target tracking, trading strict statistical optimality for association stability under this sparse regime — not an ad hoc workaround.

6.3 Mahalanobis Gating Under Saturated Covariance

The gating strategy of Section 3.3 inherits the limitation discussed by Altendorfer and Wirkert [6]: once covariance saturates at the ceiling, the gate’s acceptance region is at its widest and most permissive for the longest stretches of the recording — precisely when the filter’s true uncertainty is least well characterized by a fixed ceiling value. We did not observe incorrect cross-robot association in the three recorded scenarios, but this is a property of our specific robot spacing and motion patterns, not a guarantee that holds in denser multi-robot configurations.

7 Future Work

The continuous-error finding of Section 5 points to concrete directions rather than closing the question. **Adaptive process noise.** Instead of a fixed covariance ceiling, Q could be scaled as a function of time since the last successful correction, allowing P to reflect growing uncertainty during long gaps rather than saturating at a fixed value — a more principled alternative to covariance capping, at the cost of an additional hyperparameter to calibrate. **LiDAR as a third fusion source.** Each robot in our CERISE testbed already publishes 2D LiDAR scans that go unused in this work; fusing mutual LiDAR-based relative range, in the spirit of Khosravi et al. [7], could supply corrections during the exact intervals where the overhead camera loses visual contact, directly targeting the zero-coverage regime of Scenario 3. **Formal projection uncertainty.** Our observation model currently treats each YOLO detection’s 2D position as exact up to the confidence-derived R ; propagating the full projection Jacobian uncertainty (camera calibration error, ground-plane assumption) into R would make the confidence-to-noise mapping more principled. **Combining with distributed relative localization.** Dobrynin and Garcia’s [9] camera-free distributed Kalman filtering on the same TurtleBot3 platform is complementary rather than competing: combining robot-to-robot relative localization with our overhead-camera fusion could provide a second, independent correction channel structurally different from — and available during gaps in — visual coverage.

8 Conclusion

We replaced a discrete visual/odometry fallback in a multi-robot digital twin with an Extended Kalman Filter, validated in two stages and evaluated on real recorded data from three TurtleBot3 robots. The filter delivers a genuine, statistically significant gain at correction instants (+23.7%), but is measurably worse than raw odometry when evaluated continuously (−12.7%) under the sparse visual coverage (0–28%) our fixed overhead camera exhibits in practice — including a zero-coverage scenario where the filter’s behavior collapses, exactly as expected, to the unfused baseline. Rather than report only the favorable metric, we characterize both and ground the reversal in established theory on Kalman filtering under intermittent observations. EKF fusion remains the right tool for this problem, but its correctness under sparse observation must be measured and reported explicitly — not assumed from a single favorable number — and the low observation rate itself, not the choice of filter, is the binding constraint future work should target.

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